

handled in software by the Hadoop Framework. It is considered as the base/core of the Hadoop framework because it provides basic processes



and essential services such as abstraction of the operating system and its file system. It also contains the necessary JAR files

and scripts required to start Hadoop. It also provides source code and documentation from the Hadoop Community.

- ◆ **Hadoop YARN:** YARN is an acronym for Yet Another Resource Negotiator. YARN is a cluster management technology. YARN makes the Hadoop environment more suitable for priority based operational applications that can't wait for batch jobs to finish. It provides a central platform and resource management to deliver consistent operations, data governance tools and security across Hadoop clusters. YARN extends the power of Hadoop to adapt new technologies found within the data center so that they can take advantage of linear-scale storage, processing and cost effectiveness.
- ◆ **Hadoop Distributed File System (HDFS™):** HDFS is a Java-based file system that provides reliable and scalable storage of data, and it was designed to span large groups of commodity servers. Hadoop Distributed File System has demonstrated production scalability of up to 200 Petabytes of storage and a single cluster of 4500 servers, supporting more than a billion files and blocks. When that quantity and quality of enterprise data is available in Hadoop Distributed File System, and YARN enables multiple data access applications to process it, Hadoop users can answer the questions that jeopardized previous data platforms.

The Hadoop Distributed File System (HDFS) is based on the GFS (Google File System) and provides a distributed file system to run on large clusters of small computer machines in a reliable, fault-tolerant manner.

HDFS is a distributed storage system, fault-tolerant and scalable and works closely with a large number of concurrent data access applications, coordinated by YARN. HDFS works under a large number of physical and systemic circumstances. It distributes storage and computation across many servers and through this the combined storage resource can grow in a linear fashion with demand while remaining economical at every amount of storage.